

Part Two: Factor Analysis and Structural Equation Modeling

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Table of contents

1	What Part Two Is and Who It Is For	1
2	How the Chapters Build on One Another	3
3	How to Read a Chapter and Use the Concept Reference	8
4	What Comes Later	9

1 What Part Two Is and Who It Is For

Part Two of *Quantitative Methods for Social and Behavioral Research* is about latent-variable modeling. The methods of Part One worked with variables that were recorded: a test score, a count, a group label. Here the variable of primary interest is one that no column of the data set holds. Twelve questionnaire items are recorded; the planning, the learning strategies and the sense of belonging those items are meant to register are not. Part Two asks what follows from taking that situation seriously, and answers through factor analysis: how observed responses represent latent variables, and how a measurement model is developed, specified, estimated, evaluated and extended.

It is written for postgraduate students and applied researchers who have read Part One or who already command the statistics it develops: variable form, variance, covariance and correlation; estimates, sampling uncertainty, intervals and tests, with the limits of a dichotomous decision; regression equations, residuals and conditional comparisons; nested model comparison and penalized in-sample criteria; held-out evidence; and reproducible work in R. Part One's argument-based reading of a study is useful background rather than a prerequisite.

Four chapters form the measurement sequence, and this introduction is about them. Chapter 1, *From Latent-Variable Modeling to Factor Analysis*, is the general and conceptual presentation. Chapter 2, *Exploratory Factor Analysis and Model Development*, is the exploratory approach,

conceptual and practical. Chapter 3, *Specifying and Estimating Confirmatory Factor Models*, is the foundational confirmatory approach, covering concepts, specification and estimation. Chapter 4, *Evaluating and Comparing Confirmatory Factor Models*, is the advanced confirmatory approach, covering evaluation, comparison and the higher-order and bifactor extensions. The chapters after them carry the structural half of the subject, from relations between factors and indicators to relations among variables in a structural system; the last section here says what they cover and what they cannot confer.

Part Two asks for more mathematics than Part One did. Readers manipulate equations, count the moments a covariance matrix supplies against the parameters a model spends, and work with small vectors and matrices: a matrix product, a transpose, a symmetric matrix, a diagonal matrix. No previous course in factor algebra or matrix calculus is assumed; each new operation appears first in a scalar or two-variable case, substantive and graphical meaning comes before notation, and every symbol, index and dimension is defined before it is used. Calculus and proof-based mathematical statistics are not prerequisites, and the longer derivations, identification theorems and asymptotic arguments are placed so that a reader can pass over them without losing a definition, an equation or an assumption the main route needs.

The situation the four chapters share can be put in one sentence and one equation. Several recorded responses vary together, and the attribute used to explain their association is not recorded at all; the common-factor model writes each recorded response as a common part, shared with the other responses through a small number of unobserved variables, plus a residual part that the declared common variables do not represent. For p indicators and m factors, with $\mathbf{y}_i$ the $p \times 1$ vector of responses recorded for unit i ,

$$\mathbf{y}_i = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_i + \boldsymbol{\epsilon}_i, \quad \begin{aligned} \boldsymbol{\mu}(\boldsymbol{\vartheta}) &= \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\alpha}, \\ \boldsymbol{\Sigma}(\boldsymbol{\vartheta}) &= \boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}, \end{aligned}$$

with $E(\boldsymbol{\epsilon}_i) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\eta}_i, \boldsymbol{\epsilon}_i) = \mathbf{0}$. Here $\boldsymbol{\nu}$ holds the p indicator intercepts, $\boldsymbol{\eta}_i$ the m factors for unit i , $\boldsymbol{\Lambda}$ the $p \times m$ loadings that relate the two, and $\boldsymbol{\epsilon}_i$ the p residuals; $\boldsymbol{\alpha}$ holds the m latent intercepts, $\boldsymbol{\Phi}$ is the $m \times m$ factor covariance matrix, $\boldsymbol{\Theta}$ the $p \times p$ residual covariance matrix, and $\boldsymbol{\vartheta}$ the complete vector of free parameters. What the model produces is the implied pair: a mean structure $\boldsymbol{\mu}(\boldsymbol{\vartheta})$ and a covariance structure $\boldsymbol{\Sigma}(\boldsymbol{\vartheta})$. An analysis working from covariances alone sets the mean structure aside or profiles it out under stated assumptions, which is a choice rather than an absence. One notation serves all four chapters, and it is chosen to survive the move to structural models, so that later chapters extend the model rather than quietly changing what its symbols mean.

The covariance equation is what makes the proposal answerable. Declared loadings, factor relations and residual restrictions imply one particular covariance matrix, and that matrix can be held against the one a sample supplies. The comparison has force because a covariance matrix for p indicators holds only $p(p+1)/2$ distinct entries: a model that spends fewer parameters than that makes claims which can fail. That counting is a recurring operation

in Part Two, and it is kept separate from the further question of whether those moments determine the parameters at all.

None of this makes a factor identical to the construct that names it. A **construct** is a substantive idea belonging to the research question, and its meaning comes from theory, item content, response processes, the population studied and the use intended for the results, none of which a covariance matrix supplies. A **factor** is a random variable inside a statistical model, and an observed or estimated **score** is neither; the three are never treated as synonyms. The arrow a path diagram draws from a factor to an indicator states the direction of representation the model adopts, writing each response as a function of the factors plus a residual. It is not a finding that the factor produces the response, and it is the wrong direction for a construct whose indicators produce it.

The scope is narrower than the vocabulary might suggest. Part Two models continuous indicators and continuous outcomes; ordered categorical indicators, thresholds, polychoric correlations and ordered-response factor models are signposted where a reader would meet them and are taught elsewhere, and numeric category codes are not automatically continuous measurements. The boundary concerns the indicators rather than every variable in a study: categorical grouping variables and dummy-coded predictors stay in scope, and the later multigroup and MIMIC work uses them alongside continuous indicators. The family modelled here is the reflective common-factor model with its higher-order and bifactor relatives, estimated by normal-theory maximum likelihood across exploratory and confirmatory work; principal components analysis answers a different question and is not a common-factor estimator. What the family cannot do is as settled as what it can. No factor analysis establishes that a latent entity exists, that a reflective direction is causal, that a scale is valid, or that a structural claim is warranted; higher-order and bifactor models are rival representations of an adequate core model rather than repairs for a weak one; and reliability follows a defined score, model, population and use, so it is not validity and does not mend a weak measurement representation.

2 How the Chapters Build on One Another

The four chapters make four connected moves on one covariance equation: **represent** a common-factor model and derive the moments it implies; **explore** a development record to settle dimensionality, orientation and interpretation; **specify and estimate** the resulting proposal as explicit restrictions on evidence that played no part in producing it; and **evaluate, compare and extend** the fitted model. The equation is the same throughout. What changes at each stage is the responsibility it carries and the evidence it must answer to. Figure 1 sets out that architecture; no chapter redraws it.

Table 1 states what each chapter is responsible for and what it hands to the chapter after it.

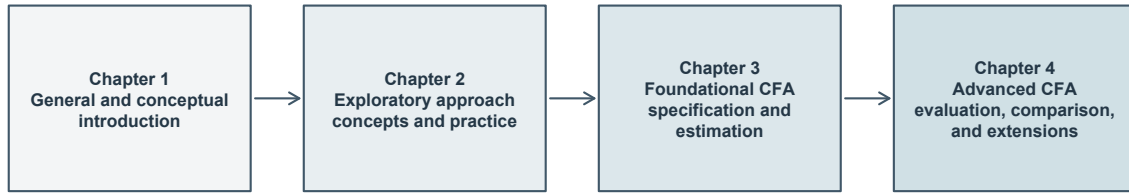


Figure 1: The four-chapter architecture moves from representation to exploratory development, foundational CFA, and advanced CFA.

Table 1

The four-chapter measurement sequence.

Chapter	Main responsibility	What it hands forward
1. From Latent-Variable Modeling to Factor Analysis	Separate an observed variable, a latent random variable, a fixed parameter and a residual, and a factor from the construct that names it; write one two-factor, six-indicator system as scalar equations, matrices, a path diagram and a sentence and check the four against one another; say what a scaling, sign, location or coordinate convention supplies and what it cannot decide; derive $\Sigma = \Lambda\Phi\Lambda^T + \Theta$; and say what exact reproduction does not establish	The vocabulary the three later chapters use without redefinition, and the known-value six-indicator records on which the algebra is checked before any sample is read

Chapter	Main responsibility	What it hands forward
2. Exploratory Factor Analysis and Model Development	Audit indicators, units, coding and missingness before reading a correlation matrix; state which matrix an extraction targets; compare observed eigenvalues with a matched permutation reference and read a structured retention profile across seven rules; count effective parameters and candidate degrees of freedom; transform loadings and factor covariance together and say what rotation preserves; and read pattern coefficients, structure correlations, factor correlations and communalities together, locating discrepancy in a residual matrix rather than an average	A frozen candidate specification with its named rivals, its retained cross-loadings, the retention evidence behind it and an explicit list of unresolved alternatives, settled before any confirmation evidence is inspected
3. Specifying and Estimating Confirmatory Factor Models	State one restricted model as words, equations, parameter matrices and a diagram and check that they agree; separate free, fixed, equality-constrained, bounded and defined entries; set a latent metric by marker or fixed-factor scaling and transform between them; trace a cross-loading and a residual covariance into the moments each changes; count moments, free parameters and degrees of freedom and say why the count is not identification; and check convergence, admissibility, gradients and local rank before reading any coefficient	An evaluation record built in memory and never written to disk: the exact specification and parameter ledger, the active items in order, the moment and degrees-of-freedom audit, the estimator and scaling settings, the convergence and admissibility record, the estimates with their uncertainty, and the rivals declared in advance

Chapter	Main responsibility	What it hands forward
4. Evaluating and Comparing Confirmatory Factor Models	Reconstruct the discrepancy the estimator minimized under a named divisor convention; read the exact-fit test, RMSEA with its interval, CFI, TLI and SRMR as answers to different questions; separate the three residual scales; establish exact nesting, run a compatible difference test and say when AIC and BIC are eligible; specify, count, fit and compare the higher-order and bifactor extensions; read modification indices and expected parameter change as leads that change analytic status; and freeze one revision and evaluate it on untouched records	A calibrated measurement conclusion whose analytic history and unresolved strain stay intact, and the structural question the later chapters take up: which latent variables are endogenous, which directional restrictions enter, and how measurement uncertainty is carried forward

Using one model family repeatedly does not give every record the same inferential role. Part Two keeps the analysis history visible so that development, first evaluation and independent revalidation are not collapsed into one favorable account. Every decision is inherited, prespecified, a planned sensitivity, or post hoc, always in relation to a named body of evidence, and that status follows the timing and provenance of the decision rather than the name of the function that fitted it: a model developed after inspecting a record is exploratory with respect to that record whatever routine estimates it, and a record made exploratory by revision does not recover confirmatory status because the next command is named after confirmation. The samples are never pooled to manufacture a larger analysis, and availability is not permission to use a record early. Table 2 gives the principal roles.

Table 2

Records keep distinct analytical roles even when they illustrate the same model family.

Record	First permitted use	What it cannot establish by itself
Known-value six-indicator records	Chapter 1 reconstructs an implied covariance matrix entrywise from parameters declared in advance	Sampling behavior, an empirical factor count, a unique rotation, a best model, or the existence of the constructs the columns are named after
SLE Sample A, 360 cases, development	Chapter 2 audits the indicators, develops the exploratory solution, and freezes the candidate with its named rivals	Independent confirmation of any choice made while Sample A was being inspected

Record	First permitted use	What it cannot establish by itself
SLE Sample B, 360 cases, confirmation	Chapter 3 estimates the frozen candidate; Chapter 4 evaluates it, diagnoses it, and compares the rivals declared in advance	Independent support for a relation first noticed in Sample B, which stays exploratory until fresh evidence tests it
SLE Sample C, 360 cases, revalidation	Chapter 4 evaluates the one revision frozen after Sample B	General validity, measurement invariance, or permission to search Sample C repeatedly for a revision that survives
Known SLE generating moments	Chapter 4 discloses them only after the Sample B and Sample C sequence is complete, then computes the population optimum and the population discrepancies directly	Contemporaneous justification for an earlier choice, or a finite-sample conclusion from one realization beyond what the audit separates
Separate six-indicator examples	Compact demonstrations of a specified calculation or model comparison, declared from their own population and seed	The development or confirmation status of the SLE sequence, which they never join

The Student Learning Experiences (SLE) records are a synthetic course case: continuous simulated data, not observations from actual students and not a validated instrument. Three independent samples of 360 complete cases carry twelve indicators – four Planning, four Strategy and four Belonging – drawn from one declared population and distributed together in `sle_continuous.csv` beside the chapters. Each analysis uses only the sample its role entitles it to, Chapter 1 uses none of them, and the Chapter 2 appendix supplies the codebook, the sample manifest and the generating population, so that every reported value can be reproduced. The retrospective audit that closes Chapter 4 is possible only because that population is known, and it is disclosed only after the Sample B and Sample C sequence has run its course: first the evidence a researcher could actually have had, then the generating structure the candidate omitted. It adds no further sample, no new candidate, no successful revision and no after-the-fact justification for a choice made earlier; its value is that a reader watches a model stay coherent, reproducible and apparently well fitting while still owing a narrower conclusion than its name suggests – which no real study can show.

The stopping points are as deliberate as the moves. Chapter 2 does not relabel a developed pattern as confirmation. Chapter 3 does not treat convergence or admissibility as adequate fit, and says so in its own section title: it reads an estimated model *before its fit*. Chapter 4 does not treat favorable fit as construct validation. Each chapter completes one responsibility and transfers a defined record, rather than carrying an unexamined conclusion forward under a new name.

3 How to Read a Chapter and Use the Concept Reference

Each chapter opens in the same way. A **Chapter overview** box summarizes the whole chapter in ordinary words: the route, and what a reader will be able to do at the end. The prose that follows it begins from a real measurement problem or concern in an applied study and states the question the chapter answers. Each chapter closes with a numbered **Summary and reading bridge** that gathers what has been settled and states the questions the next chapter owes. A short **Key terms** table follows where a chapter carries one, then exercises, then any appendices, which hold derivations, data documentation and extensions the main route does not require. A term is explained where it is first used, and the chapters carry the statistics themselves: a model appears as equations, as parameter matrices, as a path diagram and as an explanation, and every reported number is computed for the reader and read in prose and in named tables. Results are rounded to three decimals from full-precision computation, with trailing zeros dropped (3.1 rather than 3.100), unless a source's own precision applies. Counts and degrees of freedom stay whole numbers, values shown together share one number of decimals, and a very small probability is shown as a bound, such as below .001, rather than as zero. Chapter 1's exactly constructed values appear at their natural precision, and a few values carry the digits their purpose needs: a very small numerical check is written in scientific notation, and Chapter 4 shows its reconstructed discrepancy to six decimals and its population discrepancy to seven significant digits, so that the arithmetic can be followed.

The software is named where the methods need it rather than taught as a course of its own. Chapter 1 asks for none: its records are constructed, its results exact, and its conclusions readable without a computer. **psych** and **GPArotation** enter with the exploratory work of Chapter 2, and **lavaan** enters with the confirmatory fits of Chapter 3. A chapter names the software it relies on and states the consequential defaults in words – scale setting, factor covariances, mean structures, missing-data handling, standard errors and the test-statistic family – so that a reader who never opens R can still follow every model, every count and every conclusion from the printed evidence.

Part One's argument vocabulary is reused rather than redefined, and Part Two does not re-teach the argument-based approach as a recurring apparatus. One optional section of Chapter 1 shows how evidence, justification, interpretation and claim boundaries change when a statistical variable is latent, applies the six quality-evaluation questions under Part One's names, records at least one failed verdict and repairs the claim; a reader may skip it without losing a definition, an equation, an exercise or a prerequisite for the chapters that follow. Chapters 2 to 4 express prespecification, uncertainty, limitations and claim boundaries in the ordinary technical vocabulary of exploratory and confirmatory factor analysis, and *analytic status* is the one Part Two term that carries the development, confirmation and replication rules set out above.

The *Part Two Concept Reference* publishes beside these chapters and holds the single vocabulary the four of them share. Its four concept tables gather related terms into broad groups and give each term's meaning or essential distinction, and a closing table records the notation; each

row names the chapter and section where the term is principally explained, so a term met in one chapter can be traced to the place that teaches it. It is written for lookup rather than for reading through: consult it when a term or a symbol is unfamiliar, and return to the chapter for the explanation, the worked example and the reason behind the distinction.

4 What Comes Later

The chapters after Chapter 4 continue Part Two from relations between factors and indicators to relations among variables in a structural system. Their subject is observed-variable path analysis; general structural equation modeling; measurement invariance and multigroup measurement; MIMIC models; latent regression and indirect effects; latent growth in a structural equation form; and structural model comparison, ordered by their research questions rather than by a software menu. Multilevel modeling, including multilevel growth, belongs to Part Three – a division of the curriculum, not a claim that structural and multilevel parameterizations stop being equivalent when observations are unbalanced or incomplete. Item-response theory, mixture models, multilevel factor analysis and a full treatment of categorical indicators lie outside Part Two altogether.

The measurement sequence ends where a reader can read and evaluate a measurement representation, and structural work begins where a new question requires relations among variables or among factors. An acceptable measurement model is necessary evidence for many latent-variable structural analyses, but a measurement portion plus a structural portion confers no substantive warrant: structural equation modeling does not upgrade an associational design to a causal one, does not erase measurement uncertainty, and does not make an indirect effect causal by calling it mediation. Before latent means or structural parameters are compared across groups, settings or occasions, the relevant level of measurement comparability is addressed with the loading, intercept and other restrictions it requires; and a growth model addresses both mean trajectories and the variation and covariation in trajectories.

Chapter 4 therefore leaves four questions to that continuation: which latent variables are endogenous; which directional restrictions enter; whether the measurement representation is stable for the populations, groups, occasions and comparisons a structural question requires; and how measurement uncertainty is carried forward rather than replaced by exported scores. The bridge out of Chapter 4 is written with those questions in view, so that a latent intercept is not mistaken for the marginal latent mean once structural regressions enter.